R S ABISHEK

abishekrz2004@gmail.com | +91 9488968119 | Salem, Tamil Nadu

PROFESSIONAL SUMMARY

IT graduate with hands-on experience in Linux/Windows environments, Python scripting, and large-scale data processing. Worked on large-scale scientific data processing at LEOS-ISRO and developed automated monitoring and detection solutions at VRNC. Strong foundation in data preprocessing, automation, database-backed backend development (PostgreSQL/MySQL), REST APIs, Git/GitHub, and cybersecurity fundamentals.

TECHNICAL SKILLS

Systems / Other: Linux, Windows, Cybersecurity fundamentals
Programming / Scripting: Python, SQL, C, Java, JavaScript
Databases: PostgreSQL, MySQL
Tools / Platforms: Git/GitHub, Docker, Jupyter Notebook, Google Colab, VS Code
Web / APIs: Flask, Node.js, Express.js, REST API development & testing, HTML/CSS
Machine Learning / Data: PyTorch, Scikit-learn, YOLOv8, OpenCV, Pandas, NumPy, NLTK, spaCy

EXPERIENCE

Machine Learning Intern, *VRNC* 15/04/2025 – 31/05/2025 | Bangalore

- Built an automated multi-class detection system to monitor workplace safety compliance issues, achieving 85% overall accuracy.
- Analyzed 100 test images to diagnose class-confidence bias and identified a gap between majority and minority classes, applying root-cause style analysis to the model's error patterns.
- Improved minority-class recall by 20% using data augmentation and resampling to address class imbalance.
- Built an image-processing dashboard integrating the trained model with OpenCV for automated inference and continuous compliance flagging/monitoring.

Data Science Intern, *LEOS – ISRO* 15/04/2024 – 31/05/2024 | Bangalore

- Designed and automated a data preprocessing pipeline in Python for 25 million rows of space-science data using Pandas and NumPy, handling missing and erroneous timestamped entries.
- Applied pattern-recognition techniques to clean and standardize large-scale scientific datasets, improving data reliability.
- Assisted in developing software to convert space-science data into the standard PDS4 format, supporting large-scale infrastructure data operations.

EDUCATION

B.Tech in Information Technology, *CHRIST (Deemed to be University)* 2022 – 2026
CGPA: 8.45/10

SELECTED PROJECTS

Asset Management System — *Node.js, Express.js, PostgreSQL, Sequelize, Jade*

- Developed a full-stack MVC asset-management application for managing employees, assets, categories, and asset issues.
- Implemented backend routes, controllers, database models, and PostgreSQL persistence using Sequelize ORM.

Object Anomaly & Safety Compliance Detection — *Python, YOLOv8, OpenCV, Roboflow, Flask*

- Developed a computer-vision application for object anomaly and safety-compliance detection using YOLOv8.
- Integrated image upload, model inference, segmentation, and automated detection results through a Flask web application.

AdShield – Scam Ad Detection — *Python, NLP, BERT, Scikit-learn, Selenium, BeautifulSoup*

- Contributed to an AI-based social-media scam advertisement detection platform.
- Worked on user profiling using NLP, TF-IDF, clustering, and machine-learning techniques to identify targeted user groups and potential cyber-scam patterns.
- Built data-collection pipelines using Selenium and BeautifulSoup for social-media advertisement analysis.

HONORS & CERTIFICATION

- TN Police Hackathon 2024 — AdShield: 11th position; selected among 315 teams with an AI-based scam-ad detection platform.
- PCAP: Programming Essentials in Python, OpenEDG Python Institute.